# PATIENT CARE THEORY 2

## UNIT 11, PART 2: Hyperglycemia

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# Learning Outcomes

Upon successful completion of this unit, the student will have reliably demonstrated the ability to:

- Discuss the role of endocrine glands
- Discuss the role of hormones of the endocrine system
- List the common endocrine emergencies paramedics will encounter
- Compare endocrine emergencies in terms of sensitivity and specificity of signs and symptoms
- Contrast metabolic from structural cause of altered level of awareness
- Describe and defend your treatment strategy for endocrine emergencies

# HYPERGLYCEMIA

# Hyperglycemia

- Hyperglycemia –chronic
  - Diabetic ketoacidosis or DKA (Type 1, IDDM)
  - Hyperosmolar Hyperglycemic non-ketotic syndrome or HHNS (Type 2)
  - Patient not taking insulin
  - Chronic illness – cell resistance to insulin

# Hyperglycemia DKA

- Diabetic ketoacidosis or DKA
  - When cells can’t use the glucose for energy, fat cells begin to break down
  - These fat cells produce ketones which provide cellular energy; but also make the blood acidic

# Hyperglycemia DKA – Clinical Presentation

- CBG reads “high” or over 33 mmol/L
- Polyuria, Polydipsia, Polyphagia (the three “P”s)
- Dehydration
- General weakness, lethargy
- Weight loss (in the untreated or NYD patient)*
- Altered LOA
- N/V, Abd pain
- Hypotension, reflex tachycardia

  compensatory tachy in response to hypotension
- nocturia

  night pees
- Fruity breath
- Kussmaul respirations (deep, rapid breathing)

  CO2 compensation

# Hyperglycemia – Differential Diagnosis

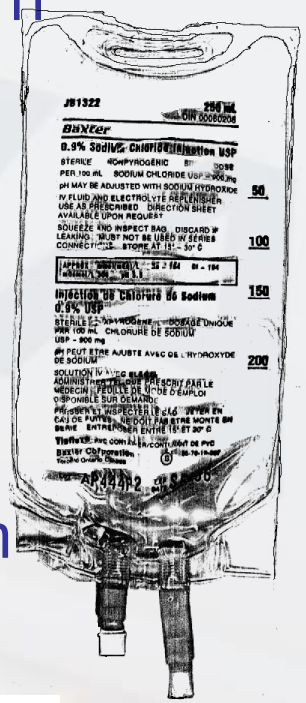
- Methanol intoxication
- Lactic acidosis (e.g. renal dialysis pt)
- Starvation acidosis
- Salicylate overdose
- Hyperosmolar hyperglycemic non-ketotic syndrome (HHNS) (AKA HHS and formerly known as HHNK Coma)

# HHNS (HHS)

- Hyperosmolar Hyperglycemic Non-ketotic syndrome OR Hyperosmolar Hyperglycemic State
  - Associated with Type 2 DM
  - Much higher mortality rate than DKA (20% vs <1%)
- Severe dehydration occurs after a period of sustained hyperglycemia. Not enough fluids have been taken to offset the hyperosmolar state and osmotic diuresis (
  - often associated with a 2° condition that prevents sufficient fluid intake
- Main symptom is altered LOA
  - S&S include many of those seen with DKA however focal or generalized seizures and transient hemiplegia may occur (which is not seen with DKA)
  - Much slower onset than DKA

# Hyperglycemia - Management

- Protect airway (NPA might be a better choice) prn
- $SpO_2$, $O_2$, PPV prn
- Cardiac monitor
- Assess CBG with glucometer (“High” = > 33 mmol/L)
- Establish IV with NS
  - Fluid bolus as per Medical Directives or patch to BHP




| Mandatory Provincial Patch Point |
| --- |
| Patch to BHP for authorization to administer 0.9% NaCl fluid bolus to hypotensive patients <12 years with suspected Diabetic Ketoacidosis (DKA). |

fluid shifts occur fast in kids, giving htem fluid will have the fluid follow sugar into brain, resulting in increased ICP and edema

# Hyperglycemia - Management

- Fluids are given and electrolytes replaced (treat the dehydration)
- Insulin do not give patient insulin. patient can take it themselves.

** blood sugar levels need to be restored to normal slowly to reduce the risk of fluid shifts within the brain

# Insulin

Common insulin:

- Very fast acting: lispro or humulog -- Onset of action 0.5-1 h, peak effect in 2-3 h; usual duration of action 8-12 h
- Fast acting: “Regular” (e.g. Humulin R, Novolin R) – peak 2-5 hours, duration 5-8 hours
- Intermediate acting: NPH* (e.g. Humulin N, Novolin N) or Lente -- Onset of action 1-1.5 h, peak effect in 4-12 h; usual duration of action 16-24 h
- Long acting: Extended insulin zinc suspension (Ultralente) -- Onset of action is 4-8 h, peak effect is in 16-18 h, and usual duration of action is >32 h.

- **Neutral Protamine Hagedorn** (NPH)
- http://www.lifeclinic.com/focus/diabetes/insulin_types.asp

# Medication for Type 2 Diabetes (NIDDM)

- Sulfonylureas: e.g. Acetohexamide (Dymelor), Chlorpropamide (Diabinese), Glimepiride (Amaryl), Glipizide (Glucotrol), **Glyburide** (**Diabeta**, Glynase, Micronase), Tolazamide (Tolinase), Tolbutamide (Orinase)
  - Forces the pancreas to produce more insulin which then lowers you BS
- Biguanides: e.g. **metformin** (Glucophage)
  - Inhibits glycogen breakdown to glucose
  - Also lowers insulin levels
  - Makes cells more sensitive to insulin
  - *Most common first medication for T2D*

ozempics and glp-1s reduce food motility, telling your brain its full (so more insulin released)

# Medication for Type 2 Diabetes (NIDDM)

- Alpha-glucosidase inhibitors: e.g. acarbose and miglitol
  - Inhibits the enzyme that digests starch → lowers BS and slows the BS rise
- Thiazolidinediones: e.g. pioglitazone (Actos) rosiglitazone (Avandia) and troglitazone (Rezulin)
  - Makes cells more sensitive to insulin
- Meglitinides: repaglinide (Prandin)
  - Stimulates the pancreas to produce more insulin
- GLP-1 receptor agonist
  - A hormone naturally produced in the gut
  - Binds to receptors to slow the movement of food
  - Act on the neurons in the brain to increase satiety
    - This reduces food intake and assists in weight loss which in turn helps with insulin resistance. Also showing cardio protective signs

# Medication for Type 2 Diabetes

- Key is understanding the effect of the medication
  - Is hypoglycemia a risk?
  - This factors in to your treatment plans and transport decisions

# QUESTIONS